Visual Correspondence Hallucination

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Problem

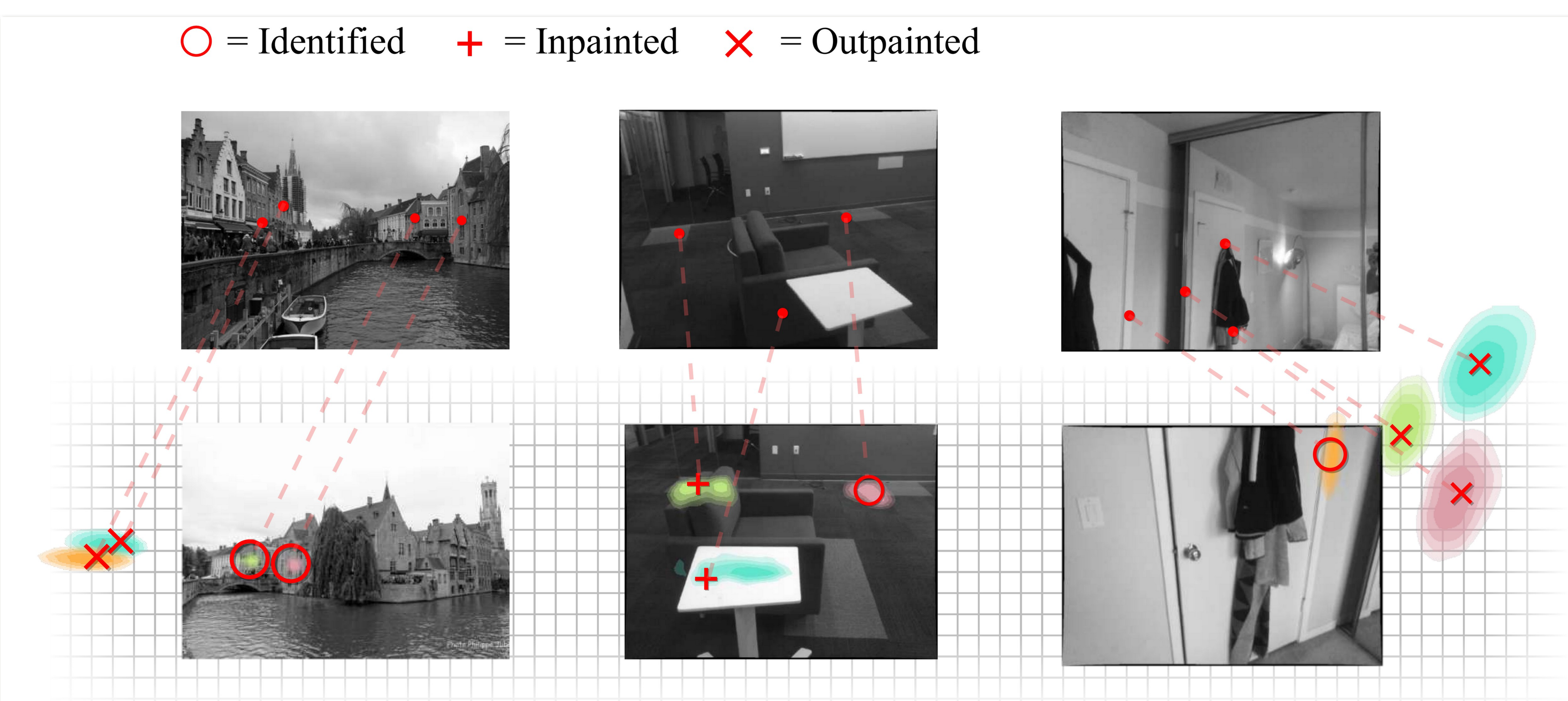
Local feature matching locates a keypoint's correspondent only when it is actually **visible** in the second image. Occluded or out-of-frame correspondents are treated as noise, so matching collapses on **low-overlap** pairs where few keypoints are covisible.

FIRST

The first method to **predict** correspondences at non-covisible locations — occluded & out-of-frame regions become signal, not noise.

Motivation

- State-of-the-art localization is built on keypoint matches, yet matchers break down precisely on the **low-overlap** pairs where most locations are occluded or outside the field of view.
- Humans instead reason about scene geometry to **hallucinate** where a non-covisible correspondent lies — a capability prior matchers lack.
- Geometry has only been used *a posteriori* (RANSAC-style outlier removal), never to **predict** non-covisible correspondents.



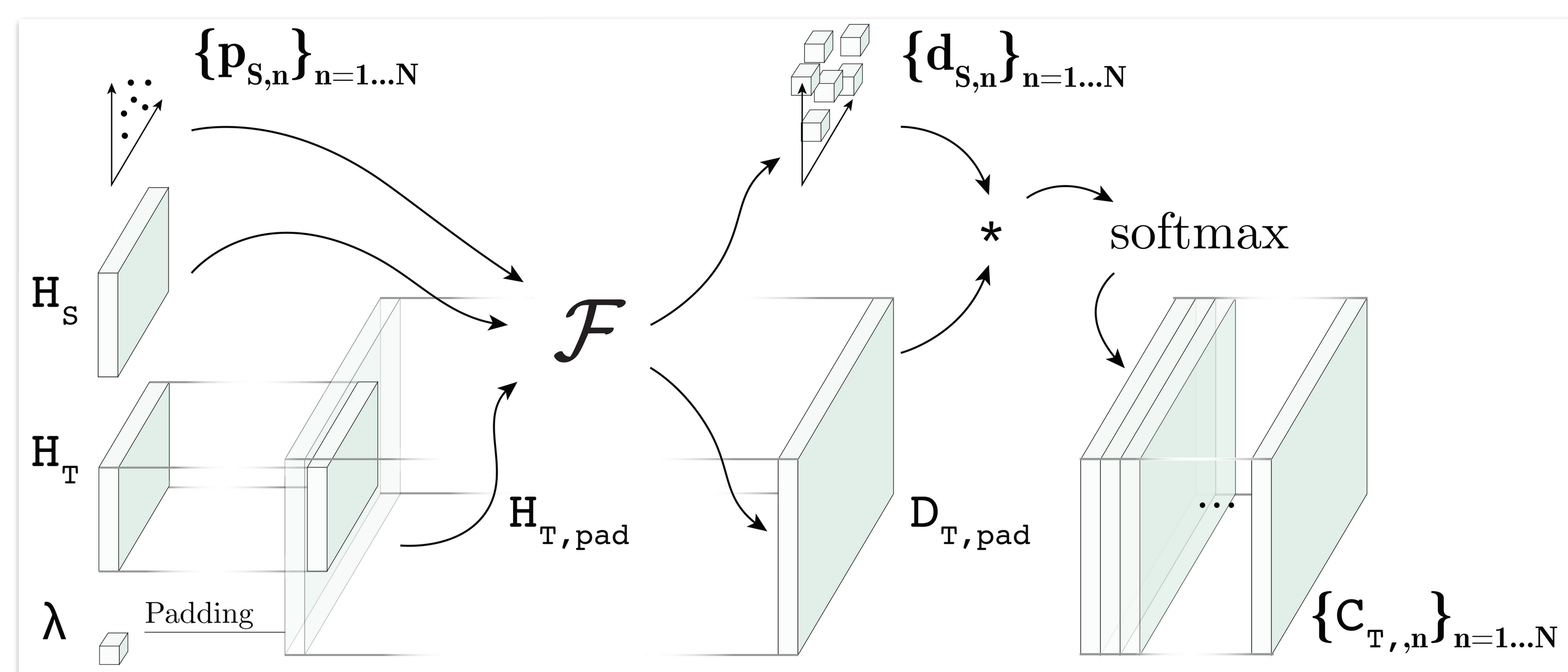
For each source keypoint, NeurHal predicts a distribution over its target correspondent — identified, inpainted, or outpainted — on unseen scenes.

Method

- A **siamese CNN** backbone produces dense descriptor maps for source and target; the target map is padded with a learnable vector **λ** to represent locations outside the field of view.
- A **cross-attention** backbone with positional encoding lets source and target descriptors communicate; each correspondence map is a 1×1 convolution of the target features by the keypoint feature, then a 2D softmax.

$$\mathrm{NRE}(p_S,\, C_T,\, R_{TS},\, t_{TS},\, d_S) = -\ln C_T(x_T)$$

reprojected location $x_T = K_C\, \omega(d_S, p_S, R_{TS}, t_{TS})$ *; training minimizes the sum of NRE terms by SGD — no covisibility labels.*



Overview of NeurHal: siamese CNN descriptors + learnable out-of-FOV padding + cross-attention yield one correspondence map per keypoint.

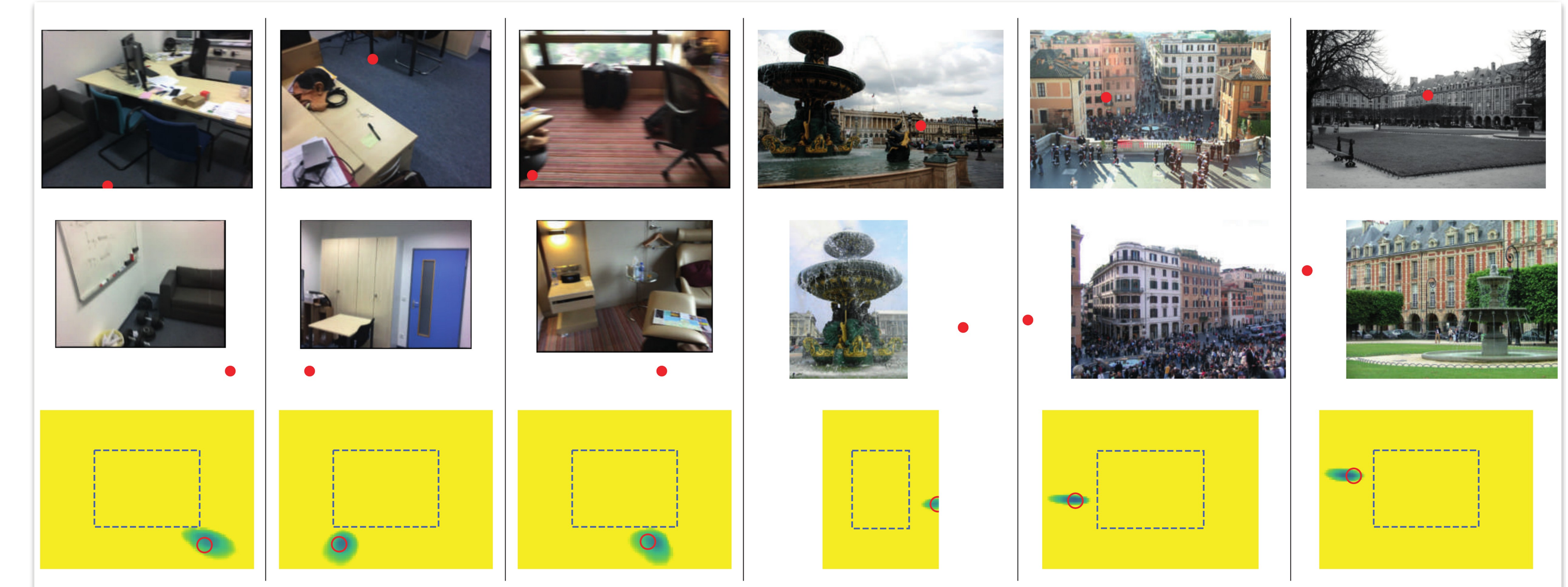
Dataset / Benchmark

- **Indoor:** trained on **ScanNet**; tested on held-out ScanNet scenes and NYU Depth.
- **Outdoor:** trained on **MegaDepth**; tested on held-out MegaDepth scenes and ETH-3D — every test scene unseen at training time.

ScanNet MegaDepth NYU Depth ETH-3D

Key Results

HALLUCINATION TASK	SOTA MATCHERS	NEURHAL
Identify · visible	✓	✓
Inpaint · occluded	poor	✓
Outpaint · out-of-frame	✗	✓



Qualitative hallucination on unseen scenes: (top) outpainting, (bottom) inpainting. Red = ground-truth correspondent; dashed box = target borders.

SO WHAT → *NeurHal's argmax errors sit far below random guessing; SOTA matchers cannot outpaint at all.*

Headline Numbers

Outpaint

correspondents beyond the target image border

the first matcher able to predict out-of-frame matches

2,500 held-out ScanNet pose pairs

<20° rotation-error threshold

<1.5m translation-error threshold

Takeaway

A single network can hallucinate keypoint correspondences — visible, occluded, or out-of-frame — making absolute camera pose estimation markedly more robust on low-overlap pairs.